Table 7.1 How challenges have shaped each generation of

Drone generation	Period/stage	Key challenges faced	Technological/strategic response
1st generation (Early UAVs)	Pre-1970s	• Limited flight stability and control accuracy • Short range and endurance • Lack of sensors and automation	• Introduction of radio-controlled systems • Basic aerodynamic and propulsion improvements
2nd generation (remote-controlled systems)	1970s–1990s	• Unreliable communication links • Limited endurance and payload • Inadequate data transmission capability	• Incorporation of GPS and telemetry • Improved remote control and navigation systems
3rd generation (semi-autonomous drones)	1990s–2010s	• High dependence on	• Integration of LiDAR, infrared,